Avances_proyecto-final

Avita Benitez Rodrigo Abdiel & Barraza Torres Jesús Daniel

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Análisis Estadístico de la Encuesta

Proyecto Final Probabilidad y Estadística

Lecturas de paquetes y conjunción de datos

```
paquetes <-
c("readxl","dplyr","janitor","knitr","ggplot2","scales","VennDiagram","ca
r")
instalados <- paquetes %in% installed.packages()[,"Package"]
if(any(!instalados)) install.packages(paquetes[!instalados],
repos="https://cloud.r-project.org")
lapply(paquetes, library, character.only = TRUE)
```

```
##
## Adjuntando el paquete: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

##
## Adjuntando el paquete: 'janitor'

## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test

## Cargando paquete requerido: grid

## Cargando paquete requerido: futile.logger

## Cargando paquete requerido: carData

##
## Adjuntando el paquete: 'car'

## The following object is masked from 'package:VennDiagram':
##
##     ellipse
```

```
## The following object is masked from 'package:dplyr':
##
##     recode
```

```
## [[1]]
## [1] "readxl"    "stats"     "graphics"  "grDevices" "utils"
"datasets"
## [7] "methods"   "base"
##
## [[2]]
## [1] "dplyr"     "readxl"    "stats"     "graphics"  "grDevices"
"utils"
## [7] "datasets"  "methods"   "base"
##
## [[3]]
##  [1] "janitor"   "dplyr"     "readxl"    "stats"     "graphics"
"grDevices"
##  [7] "utils"     "datasets"  "methods"   "base"
##
## [[4]]
##  [1] "knitr"     "janitor"   "dplyr"     "readxl"    "stats"
"graphics"
##  [7] "grDevices" "utils"     "datasets"  "methods"   "base"
##
## [[5]]
##  [1] "ggplot2"   "knitr"     "janitor"   "dplyr"     "readxl"
"stats"
##  [7] "graphics"  "grDevices" "utils"     "datasets"  "methods"
"base"
##
## [[6]]
##  [1] "scales"    "ggplot2"   "knitr"     "janitor"   "dplyr"
"readxl"
##  [7] "stats"     "graphics"  "grDevices" "utils"     "datasets"
"methods"
## [13] "base"
##
## [[7]]
##  [1] "VennDiagram"   "futile.logger" "grid"          "scales"
##  [5] "ggplot2"       "knitr"         "janitor"       "dplyr"
##  [9] "readxl"        "stats"         "graphics"      "grDevices"
## [13] "utils"         "datasets"      "methods"       "base"
##
## [[8]]
##  [1] "car"           "carData"       "VennDiagram"   "futile.logger"
##  [5] "grid"          "scales"        "ggplot2"       "knitr"
##  [9] "janitor"       "dplyr"         "readxl"        "stats"
## [13] "graphics"      "grDevices"     "utils"         "datasets"
## [17] "methods"       "base"
```

```r
if (requireNamespace("rstudioapi", quietly = TRUE) &&
rstudioapi::hasFun("selectFile")) {
  ruta_excel <- rstudioapi::selectFile(caption = "Selecciona el archivo
de la encuesta",
                                       filter = "Excel Files (*.xlsx)")
} else {
  ruta_excel <- file.choose()
}


datos <- read_excel(ruta_excel)
datos <- clean_names(datos)

carrera <- as.factor(datos[[2]])
cuatrimestre <- as.factor(datos[[3]])
materias_reprobadas <- as.numeric(datos[[4]])
horas_estudio <- as.numeric(datos[[5]])

df <- data.frame(carrera, cuatrimestre, materias_reprobadas,
horas_estudio)
```

Frecuencias cualitativas

```r
tab_carrera <- table(df$carrera)
tab_cuatr <- table(df$cuatrimestre)

kable(as.data.frame(tab_carrera), caption="Frecuencias de Carrera")
```

Frecuencias de Carrera

Var1	Freq
Desarrollo de negocios	35
Diseño digital	13
Lengua Inglesa	4
Logística	19
Mantenimiento industrial	3
Mecatrónica	13
Tecnologías de la información	15

```r
kable(as.data.frame(tab_cuatr), caption="Frecuencias de Cuatrimestre")
```

Frecuencias de Cuatrimestre

Var1	Freq
3	32
5	29

Var1	Freq
8	39
11	2

Frecuencia para variable cuantitativa

```
x <- df$materias_reprobadas
x <- x[!is.na(x)]
n <- length(x)
k <- ceiling(1 + log2(n))
breaks <- pretty(range(x), n = k)
clases <- cut(x, breaks=breaks, right=FALSE, include.lowest=TRUE)
freq <- table(clases)
midpoints <- (head(breaks,-1)+tail(breaks,-1))/2

tabla_freq <- data.frame(
  lim_inf = head(breaks,-1),
  lim_sup = tail(breaks,-1),
  marca = midpoints,
  frecuencia = as.integer(freq),
  frec_rel = round(as.numeric(freq)/n,4)
)
kable(tabla_freq, caption="Tabla de frecuencia - Materias reprobadas")
```

Tabla de frecuencia - Materias reprobadas

lim_inf	lim_sup	marca	frecuencia	frec_rel
0	1	0.5	40	0.3922
1	2	1.5	32	0.3137
2	3	2.5	16	0.1569
3	4	3.5	8	0.0784
4	5	4.5	2	0.0196
5	6	5.5	1	0.0098
6	7	6.5	1	0.0098
7	8	7.5	1	0.0098
8	9	8.5	0	0.0000
9	10	9.5	1	0.0098

```
y <- df$horas_estudio
y <- y[!is.na(y)]
n1 <- length(y)
k1 <- ceiling(1 + log2(n1))
breaks1 <- pretty(range(y), n1 = k1)
clases1 <- cut(y, breaks=breaks1, right=FALSE, include.lowest=TRUE)
freq1 <- table(clases1)
```

```r
midpoints1 <- (head(breaks1,-1)+tail(breaks1,-1))/2

tabla_freq <- data.frame(
  lim_inf = head(breaks1,-1),
  lim_sup = tail(breaks1,-1),
  marca = midpoints1,
  frecuencia = as.integer(freq1),
  frec_rel = round(as.numeric(freq1)/n1,4)
)
kable(tabla_freq, caption="Tabla de frecuencia - Materias reprobadas")
```

Tabla de frecuencia - Materias reprobadas

lim_inf	lim_sup	marca	frecuencia	frec_rel
0	5	2.5	72	0.7059
5	10	7.5	22	0.2157
10	15	12.5	6	0.0588
15	20	17.5	2	0.0196

Medidas descriptivas

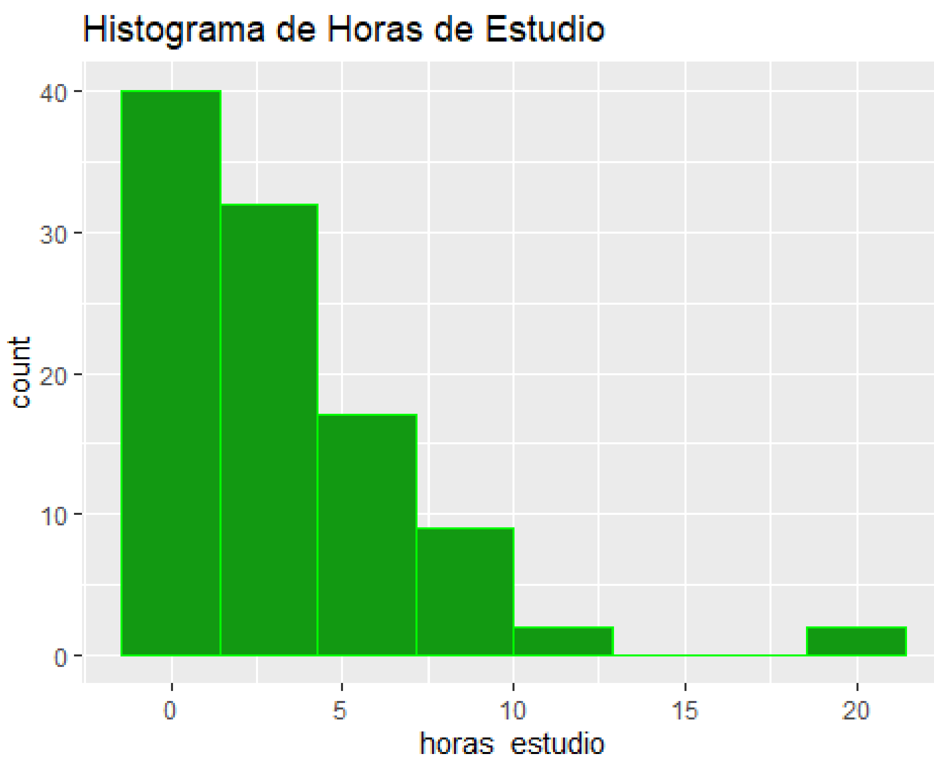
```r
resumen_var <- function(v){
  v <- v[!is.na(v)]
  media <- mean(v)
  mediana <- median(v)
  ux <- unique(v); tab <- tabulate(match(v, ux))
  moda <- ux[which(tab==max(tab))]
  rango <- diff(range(v))
  varianza <- var(v)
  desv <- sd(v)
  cuartiles <- quantile(v, c(.25,.5,1))
  deciles <- quantile(v, seq(.1,1,.1))
  percentiles <- quantile(v, seq(0.01, 1, 0.01))
  list(media=media, mediana=mediana, moda=moda, rango=rango,
       varianza=varianza, desviacion=desv, cuartiles=cuartiles,
deciles=deciles, percentiles=percentiles)
}
resumen_var(df$materias_reprobadas)

## $media
## [1] 1.215686
##
## $mediana
## [1] 1
##
## $moda
## [1] 0
##
```

```
## $rango
## [1] 10
##
## $varianza
## [1] 2.56688
##
## $desviacion
## [1] 1.602149
##
## $cuartiles
##   25%  50% 100%
##     0    1   10
##
## $deciles
##   10%  20%  30%  40%  50%  60%  70%  80%  90% 100%
##     0    0    0    1    1    1    1    2    3   10
##
## $percentiles
##    1%    2%    3%    4%    5%    6%    7%    8%    9%   10%   11%
12%   13%
##  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00
0.00  0.00
##   14%   15%   16%   17%   18%   19%   20%   21%   22%   23%   24%
25%   26%
##  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00
0.00  0.00
##   27%   28%   29%   30%   31%   32%   33%   34%   35%   36%   37%
38%   39%
##  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00
0.00  0.39
##   40%   41%   42%   43%   44%   45%   46%   47%   48%   49%   50%
51%   52%
##  1.00  1.00  1.00  1.00  1.00  1.00  1.00  1.00  1.00  1.00  1.00
1.00  1.00
##   53%   54%   55%   56%   57%   58%   59%   60%   61%   62%   63%
64%   65%
##  1.00  1.00  1.00  1.00  1.00  1.00  1.00  1.00  1.00  1.00  1.00
1.00  1.00
##   66%   67%   68%   69%   70%   71%   72%   73%   74%   75%   76%
77%   78%
##  1.00  1.00  1.00  1.00  1.00  1.71  2.00  2.00  2.00  2.00  2.00
2.00  2.00
##   79%   80%   81%   82%   83%   84%   85%   86%   87%   88%   89%
90%   91%
##  2.00  2.00  2.00  2.00  2.00  2.00  2.00  2.00  2.87  3.00  3.00
3.00  3.00
##   92%   93%   94%   95%   96%   97%   98%   99%  100%
##  3.00  3.00  3.00  3.95  4.00  4.97  5.98  6.99 10.00
```

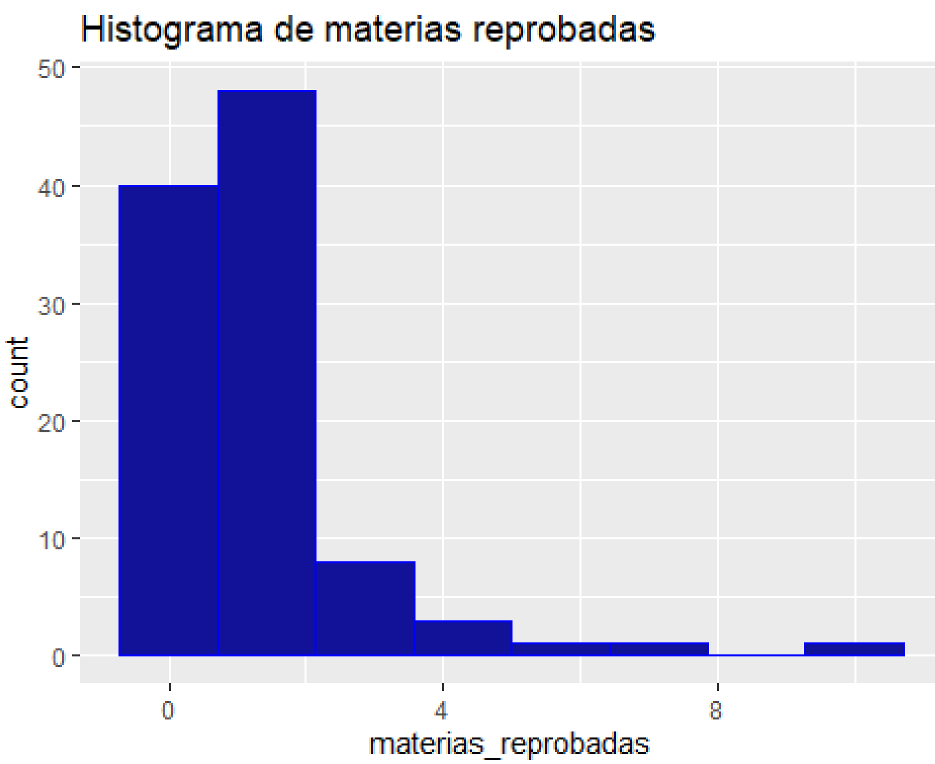
```r
resumen_var(df$horas_estudio)
```

```
## $media
## [1] 3.254902
##
## $mediana
## [1] 2
##
## $moda
## [1] 0
##
## $rango
## [1] 20
##
## $varianza
## [1] 14.90468
##
## $desviacion
## [1] 3.860658
##
## $cuartiles
##  25%  50% 100%
##    0    2   20
##
## $deciles
##  10%  20%  30%  40%  50%  60%  70%  80%  90% 100%
##    0    0    1    2    2    3    4    5    8   20
##
## $percentiles
##    1%    2%    3%    4%    5%    6%    7%    8%    9%   10%   11%
12%   13%
##  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00
0.00  0.00
##   14%   15%   16%   17%   18%   19%   20%   21%   22%   23%   24%
25%   26%
##  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00  0.00
0.00  0.00
##   27%   28%   29%   30%   31%   32%   33%   34%   35%   36%   37%
38%   39%
##  0.00  0.28  1.00  1.00  1.00  1.00  1.00  1.00  1.00  1.00  1.00
1.00  1.39
##   40%   41%   42%   43%   44%   45%   46%   47%   48%   49%   50%
51%   52%
##  2.00  2.00  2.00  2.00  2.00  2.00  2.00  2.00  2.00  2.00  2.00
2.00  2.00
##   53%   54%   55%   56%   57%   58%   59%   60%   61%   62%   63%
64%   65%
##  2.00  2.00  2.00  2.56  3.00  3.00  3.00  3.00  3.00  3.00  3.00
3.00  3.00
##   66%   67%   68%   69%   70%   71%   72%   73%   74%   75%   76%
77%   78%
##  3.00  3.67  4.00  4.00  4.00  4.71  5.00  5.00  5.00  5.00  5.00
```

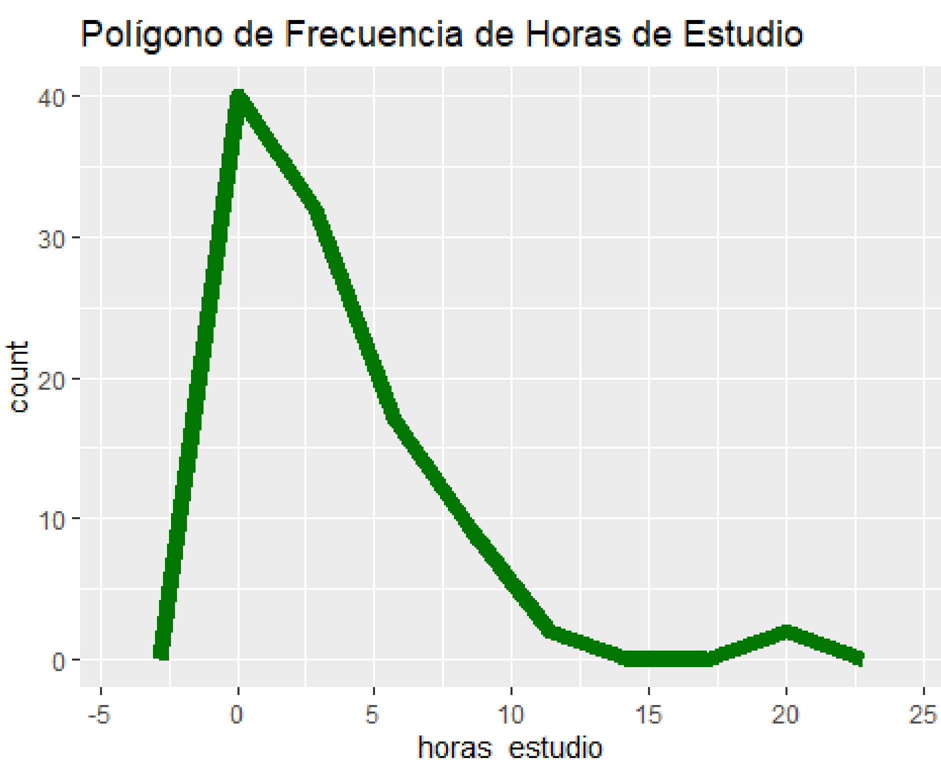
```
5.00  5.00
##    79%    80%    81%    82%    83%    84%    85%    86%    87%    88%    89%
90%   91%
##  5.00  5.00  5.00  5.00  5.83  6.00  6.85  7.00  7.00  7.88  8.00
8.00  8.00
##    92%    93%    94%    95%    96%    97%    98%    99%   100%
##  8.92  9.93 10.00 10.00 10.00 11.94 12.00 19.92 20.00

# Histogramas, polígono y ojiva
ggplot(df, aes(x=horas_estudio)) +
  geom_histogram(bins=8, color="green", fill="#129912") +
  labs(title="Histograma de Horas de Estudio")
```



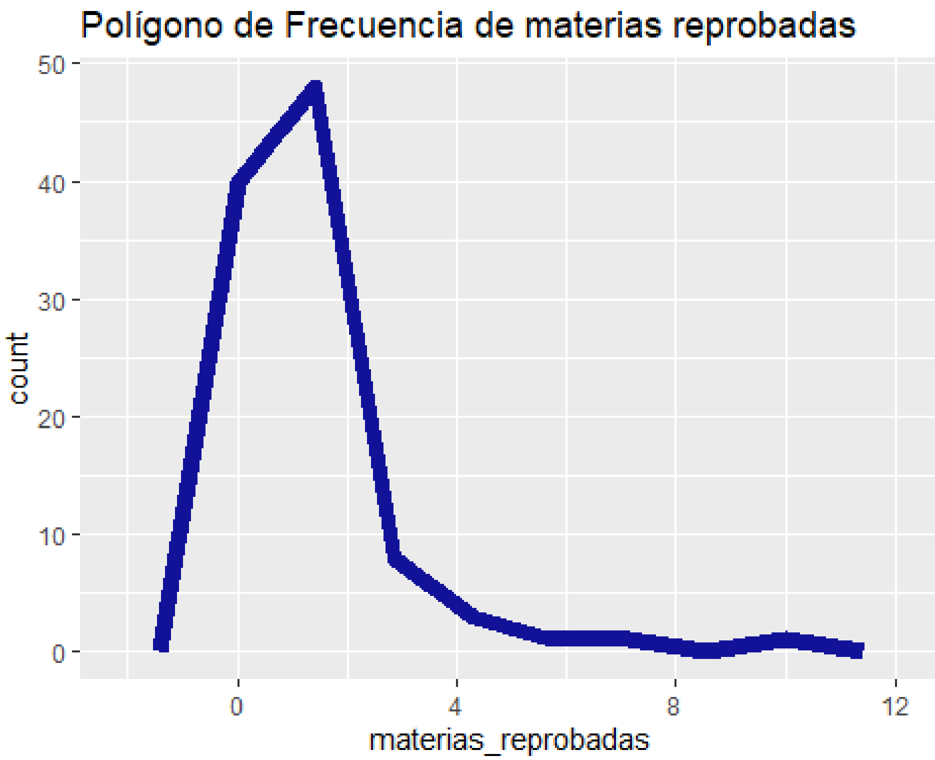

```
ggplot(df, aes(x=materias_reprobadas)) +
  geom_histogram(bins=8, color="blue", fill="#121299") +
  labs(title="Histograma de materias reprobadas")
```



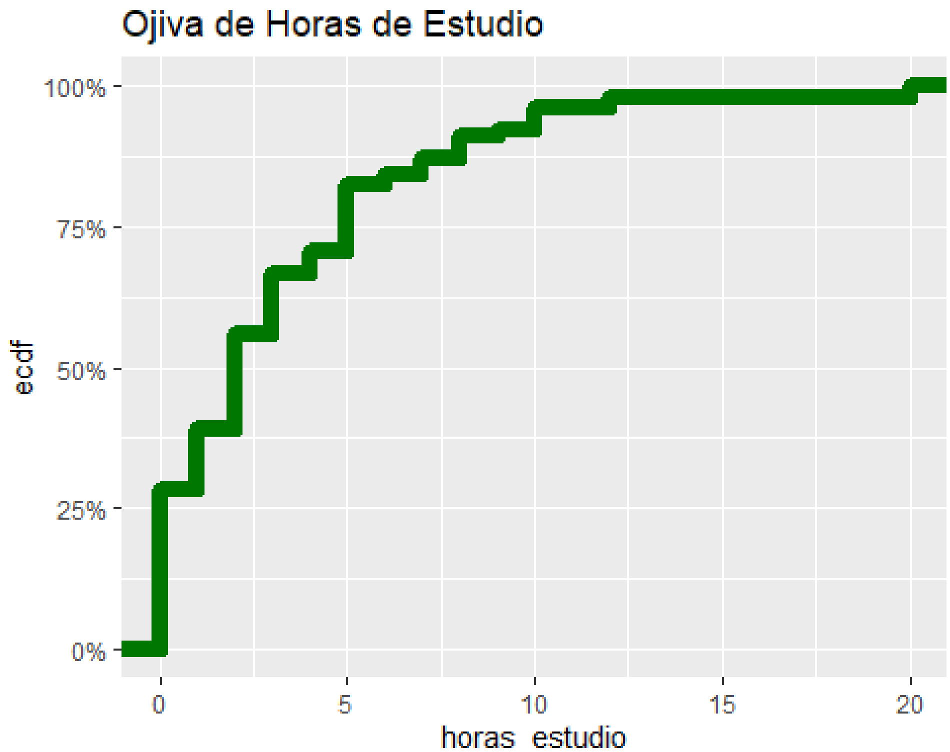

```r
ggplot(df, aes(x=horas_estudio)) +
  geom_freqpoly(bins=8, linewidth=3, color="#007700") +
  labs(title="Polígono de Frecuencia de Horas de Estudio")
```



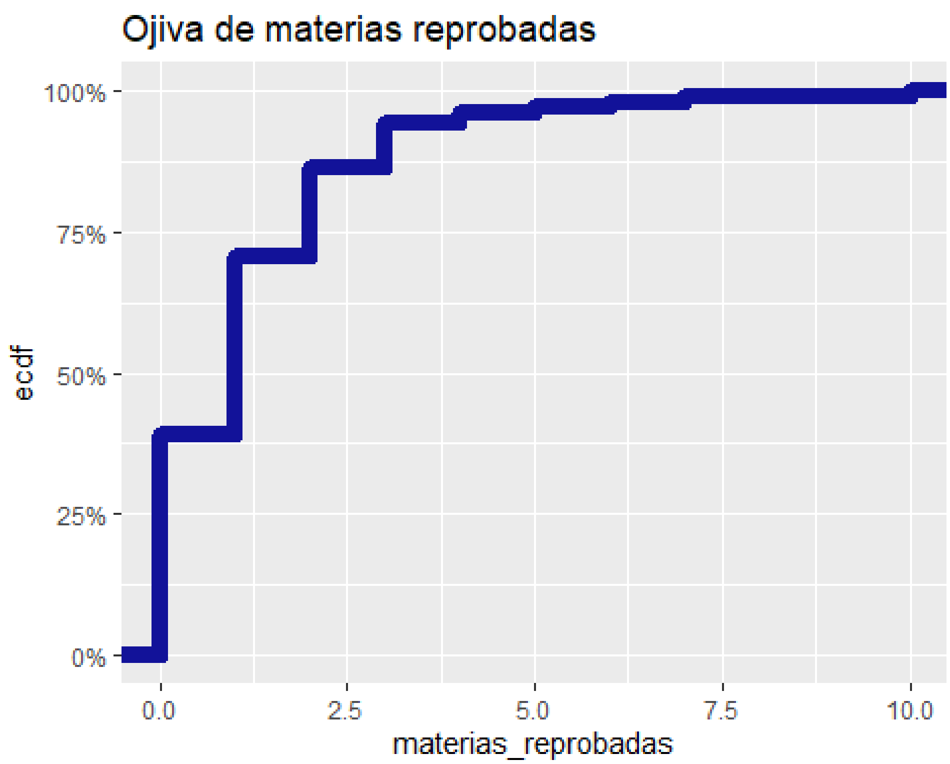
```r
ggplot(df, aes(x=materias_reprobadas)) +
  geom_freqpoly(bins=8, linewidth=3, color="#121299") +
  labs(title="Polígono de Frecuencia de materias reprobadas")
```




```r
ggplot(df, aes(x=horas_estudio)) +
  stat_ecdf(geom="step", linewidth=3, color="#007700") +
  scale_y_continuous(labels=percent_format()) +
  labs(title="Ojiva de Horas de Estudio")
```

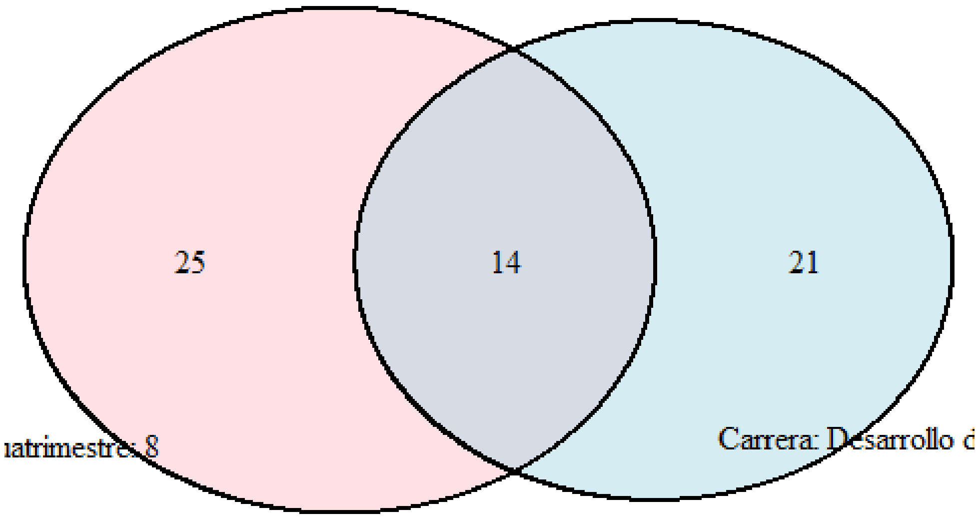



```r
ggplot(df, aes(x=materias_reprobadas)) +
  stat_ecdf(geom="step", linewidth=3, color="#121299") +
  scale_y_continuous(labels=percent_format()) +
  labs(title="Ojiva de materias reprobadas")
```



Probabilidades con diagrama de Venn

```
top_carrera <- names(sort(tab_carrera, decreasing=TRUE))[1]
top_cuatr <- names(sort(tab_cuatr, decreasing=TRUE))[1]

setA <- df$carrera==top_carrera
setB <- df$cuatrimestre==top_cuatr
nA <- sum(setA); nB <- sum(setB); nAB <- sum(setA & setB)

grid.newpage()
draw.pairwise.venn(area1=nA, area2=nB, cross.area=nAB,
                   category=c(paste("Carrera:", top_carrera),
                              paste("Cuatrimestre:", top_cuatr)),
                   fill=c("lightblue","pink"), alpha=c(.5,.5))
```




```
## (polygon[GRID.polygon.228], polygon[GRID.polygon.229],
polygon[GRID.polygon.230], polygon[GRID.polygon.231],
text[GRID.text.232], text[GRID.text.233], text[GRID.text.234],
text[GRID.text.235], text[GRID.text.236])
```

```
P_A <- nA/n; P_B <- nB/n; P_AB <- nAB/n; P_AcondB <- P_AB/P_B
c(P_A, P_B, P_AB, P_AcondB)
```

```
## [1] 0.3431373 0.3823529 0.1372549 0.3589744
```

Prueba de hipótesis

```
grupo0 <- df$horas_estudio[df$materias_reprobadas==0]
grupo1 <- df$horas_estudio[df$materias_reprobadas>0]

if(length(grupo0)>=3 & length(grupo1)>=3){
  if(shapiro.test(grupo0)$p.value>0.05 &
shapiro.test(grupo1)$p.value>0.05){
    t.test(grupo0, grupo1)
  } else {
    wilcox.test(grupo0, grupo1)
  }
}
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
```

```
## data:  grupo0 and grupo1
## W = 1238, p-value = 0.9917
## alternative hypothesis: true location shift is not equal to 0
```

Correlación y regresión

```
df_cor <- na.omit(df)
cor.test(df_cor$materias_reprobadas, df_cor$horas_estudio,
method="pearson")
```

```
##
##  Pearson's product-moment correlation
##
## data:  df_cor$materias_reprobadas and df_cor$horas_estudio
## t = 0.32659, df = 100, p-value = 0.7447
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.1628667  0.2256843
## sample estimates:
##        cor
## 0.03264207
```

```
lm_mod <- lm(horas_estudio ~ materias_reprobadas, data=df_cor)
summary(lm_mod)
```

```
##
## Call:
## lm(formula = horas_estudio ~ materias_reprobadas, data = df_cor)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -3.710 -3.159 -1.238  1.762 16.841
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)          3.15928    0.48286   6.543  2.6e-09 ***
## materias_reprobadas  0.07866    0.24084   0.327    0.745
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.878 on 100 degrees of freedom
## Multiple R-squared:  0.001066,   Adjusted R-squared:  -0.008924
## F-statistic: 0.1067 on 1 and 100 DF,  p-value: 0.7447
```

```
correlacion <- cor(df$materias_reprobadas, df$horas_estudio)
cat("The correlation between the studying time and the failed
assignatures is for: ", correlacion)
```

```
## The correlation between the studying time and the failed assignatures
is for:  0.03264207
```

```r
ggplot(df_cor, aes(x=materias_reprobadas, y=horas_estudio)) +
  geom_point() +
  geom_smooth(method="lm", se=TRUE) +
  labs(title="Horas de estudio vs Materias reprobadas")

## `geom_smooth()` using formula = 'y ~ x'
```